Homework 2: Improving RAG with Wikidata Information Retrieval: an In-Depth Analysis

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1 Introduction

In this report, we describe the implementation of a RAG pipeline in QA, which builds on the findings on the Semantic Search task from the previous homework. We built a search algorithm on Wikidata entities to bridge the retrieval gap of a small retriever like All-Mini-L6-v2, and thus improve the context for the models. To this end, we evaluate the results of a Seq2Seq and a causal model. We then motivate the results and try to answer the following question: is Wikidata context enrichment going to improve models' performance and scores on a Question Answering task?

2 Methodology

We first defined the two models which have been used for the given task. The models are as follows:

- **flan-t5-large**: a 0.8B encoder-decoder model which has been finetuned on different tasks using instruction finetuning, which consists of asking the model to answer/solve correctly a set of questions/tasks which have been phrased as instructions;
- **Llama-3.2-1B-Instruct**: a 1B decoder-only model fine-tuned for instruction-following and dialogue use cases. It also excels in knowledge retrieval and summarization;

For RAG and Oracle pipelines, the implemented document retriever is all-Mini-L6-v2. The intuition under the Wikidata information retrieval algorithm is to define a search algorithm that searches information about the query-related entities, including a maximum amount of 10 properties (relations between the main entity and other entities/properties): the result is an additional document to the ones retrieved from all-Mini-L6-v2 in the RAG pipeline, that can cover residual context when the correct answer's document is missing.

The automatic evaluation setup relies on the default metrics (EM, subEM and METEOR) computed over the short (correct) answer and a Wikidata ground truth augmentation to retrieve other synonyms or aliases with respect to the gold answer, and a LLM-as-a-Judge approach using Mistral-7B-Instruct-v0.3, which is a great performing LLM for point-wise scoring (Wang et al., 2025).

3 Experimental Setup

The Baseline pipeline is a simple implementation of the libraries necessary to perform question answering on a LLM, so we are not going to dedicate any space to deepen its structure. As for the RAG and Oracle pipelines, both rely on all-Mini-L6-v2 to retrieve the top-3 chunks by Semantic Search. Every context in RAG also includes the Wikidata information retrieval algorithm result, which plays the role of an additional retrieved document. More details can be found in the Appendix A.1

Both flan-t5-large and Llama-3.2.-1B-Instruct present the same generation parameters, which are explored more in detail in the Appendix A.2. The decision to apply identical generation parameters was made to establish a fair baseline for both: a temperature level below 1.0 forces LLaMA to be less creative; on the other hand, it gives FLAN-T5 generally more space for verbosity. We expect LLaMA to perform worse in EM scores, but we also hope for a fairer comparison between both models with respect to the subEM and the METEOR evaluation.

For the evaluation phase, the LLM-as-a-Judge setup is simple: the model does not need to predict tokens for generation, it needs to judge, so it is not allowed to sample any token (see Appendix A.2). Also, since Mistral-7B-v0.3 is a reasoning model, we considered as a best-practice to let the model generate an answer divided in two parts:

- **1st part (Reasoning)**: The model is asked to reason about the generated answer and decide whether it is correct or incorrect;
- **2nd part (Score)**: The model has to give a

binary score to the answer, based on the reasoning made above;

This bridges the evaluation problems for reasoning LLMs, where these kinds of models lack in giving a strict binary score as an answer.

4 Results

4.1 Comparison Between Pipelines

The automated evaluation over the 2000-query test set highlights a stark contrast between the two architectures across the three pipelines (see Appendix 1). The Baseline shows very poor performance for both models, reflecting the models' lack of internal knowledge to directly answer the specific queries.

However, the implementation of the RAG and Oracle pipelines induces a massive performance improvement. This convergence suggests that the Wikidata context mitigates the risks of sub-optimal retriever ranking, allowing the model to perform at a level comparable to having the gold-standard chunk prioritized, even when the retriever is forced to contend with less ideal ordering. A richer description can be read in the Appendix A.4.

4.2 Wikidata Ground Truth Enrichment

The data provides strong support for the hypothesis that structured knowledge improves semantic matching. When augmenting the ground truth with the Wikidata search algorithm, we observe a consistent score bump across all metrics for both models. More details are in the appendix A.4.

4.3 Human and LLM-as-a-Judge Annotations

We relied on Mistral-7B-Instruct-v0.3 to evaluate FLAN-T5-large on a randomly sampled set of 250 queries. The LLM Judge scored the model with a 0.7480 average correctness, while human annotators reported an average score of 0.7440.

The alignment between the LLM Judge and human evaluation is substantial, yielding a Cohen's Kappa of 0.6728. While it indicates a solid base of agreement, the metric highlights that Mistral occasionally penalized linguistically correct and semantically valid answers for not matching the short answer word-for-word. This behaviour could be highly driven by the system prompt: we leave a deeper analysis in the Appendix A.2.

4.4 Qualitative Analysis

A granular qualitative review of the sample subset highlights three primary failure modes and behavioral patterns governing the generative execution:

- **Context Order Bias:** T5 strongly relies on the first retrieved document whenever it recognizes a semantically coherent answer pattern. Interestingly, in scenarios where the top-ranked document visibly fails to answer the question, the LLM bypasses the first block and navigates through the subsequent retrieved passages to extract the correct answer.
- **Wikidata Information Retrieval Relevance:** When the retriever completely fails to fetch documents containing the gold answer, FLAN-T5 successfully exploits the raw facts brought by the Wikidata information retrieval layer. When the primary documents are irrelevant, the structured text acts as an anchor of truth, guiding the model's to a proper answer.
  Wikidata enrichment is highly effective for strict, direct-factual questions (e.g., specific attributes of a subject, locations, or clear timestamps). Conversely, when a query is non-direct or non-factual and the primary RAG documents are sub-optimal, the Wikidata search algorithm offers little to no utility. This is an intrinsic limitation of the algorithm's design: it is engineered to extract atomic relationships and entities, meaning it lacks the broader narrative synthesis and contextualized descriptions required to resolve complex, indirect questions.
- **Dataset Ambiguity Handling:** To tackle queries which are ambiguous with respect to the purpose of the answer, the model consistently relies on specific sentences or words that are precisely given from the retrieved context (both for RAG documents and Wikidata information). When retrieved passages and Wikidata facts disagree, the model often prioritizes the Wikidata information.
- **Entity Overlap:** In rare cases, the model is unable to output the correct response even when both the retrieved documents and the Wikidata context explicitly contain the solution. This is heavily driven by multiple entity overlaps, which generate severe semantic interference, causing the model to prioritize a salient distractor entity.

A comprehensive breakdown of these specific examples is detailed in Appendix A.6.

A Appendix

A.1 Wikidata Search Algorithms and Gold Properties

In this section, we provide the predefined set of properties utilized during the extraction phase, along with a list of pseudocodes which define the process of searching entities, properties, labels, and descriptions in Wikidata for the tasks described in the Experimental Setup.

To maximize the relevance of the retrieved context and filter out noisy or irrelevant metadata, we manually selected 20 high-value "Gold Properties" from the top-100 most used properties in the Wikidata knowledge base. These properties, listed in Table 2, prioritize actionable semantic relationships for QA tasks, such as temporal bounds, geographical locations, and taxonomic classifications.

Algorithm 1 outlines the extraction of basic entity information. It retrieves labels, descriptions, and aliases for a given set of Wikidata IDs, formatting them into a readable text block to be injected into the context as the "fourth document".

Algorithm 2 manages the enrichment of an entity's context. It performs a two-stage extraction from the entity's claims: first identifying high-priority "gold" properties, then filling the remaining quota (up to 10) with other properties. Finally, it resolves any retrieved IDs into human-readable labels to ensure the context is fully textualized.

Algorithm 3 details the property filtering mechanism. It limits the extraction to the top 10 relevant properties, prioritizing the predefined gold properties (see Table 2), while discarding unhelpful data types such as URLs or multimedia files.

Algorithm 4 describes the logic behind the automated metric evaluation enhancement. It cross-references the original short answer with the entity's properties to pinpoint the exact Wikidata ID of the expected answer, subsequently fetching all its valid aliases to dynamically expand the ground truth.

A.2 LLMs Parameters Setup

Generative Model Parameters The parameters of the generative models are defined as follows:

- `max_new_tokens = 50`;
- `temperature = 0.7`;
- `top_p = 0.9`;
- `do_sample = True`;
- `num_beams = 1`;

First, `max_new_tokens` parameter has been set to 50 both to limit the generation and to optimize the computational budget. The decision to fix a lower temperature for both models, as explained in the main report, has been made to create a fair baseline in the middle between the two approaches: the more creative model, llama-3.2-1B-instruct, is forced to create more pragmatic and direct answers, while flan-t5-large maintains a mild approach since it has already fine-tuned for responding with brief and precise answers. The nucleus sampling has been used to let the models focus on predicting the next token in a smaller batch of candidates: this again enforces the concept of briefness. The `num_beams` is set to 1 for the same reason: the goal is to obtain strict answers, so the beam search is not relevant in this case.

LLM-as-a-Judge Parameters The LLM-as-a-Judge model's parameters are defined as follows:

- `max_new_tokens = 150`;
- `do_sample = False`

Since the LLM-as-a-Judge model needs to provide a binary score which does not require any specific token generation, `do_sample` has been set to 0. Even though the model requires a Chain-of-Thought reasoning window, our purpose was not to rely on it, since it has been set only for a clearer score extraction from model's answer. This also justifies the `max_new_tokens` parameter set to 150, since it allows a longer Chain-of-Thought reasoning process to the model, avoiding undesired scenarios where the final scoring token would be omitted.

A.3 Prompt Setup

System Prompt (RAG)

"You are an expert in question answering. Given a set of retrieved documents, and information about a Wikidata entity, provide a concise and correct answer to the question.

Context: {context}
Query: {query} *"*

Model	Pipeline	Exact Match (EM)		Sub-string EM (subEM)		METEOR	
		Standard	Wikidata	Standard	Wikidata	Standard	Wikidata
FLAN-T5-large	Baseline	0.0245	0.0280	0.0350	0.0495	0.0650	0.0717
	RAG	0.3475	0.3785	0.4505	0.5200	0.4627	0.4873
	Oracle	0.4250	0.4490	0.5435	0.6005	0.5355	0.5547
LLaMA-3.2-1B-Instruct	Baseline	0.0020	0.0025	0.1060	0.1550	0.0960	0.1036
	RAG	0.0595	0.0670	0.3525	0.4410	0.2929	0.3093
	Oracle	0.0305	0.0325	0.3815	0.4525	0.2695	0.2804

Table 1: Automated evaluation metrics on the 2000-query test set. The *Standard* columns refer to the Exact Match with the original short answer, while the *Wikidata* columns display the scores recalculated using the Wikidata-expanded ground truth.

Algorithm 1: Retrieve Entity Information (`get_wikidata_entity`)

Input: $wikidata_ids$ (List of IDs), id_only (Boolean flag)
Output: Dictionary of labels **if** id_only **else** Formatted String
$joined_ids \leftarrow$ Join $wikidata_ids$;
$response \leftarrow$ Call API(action = "wbgetentities", ids = $joined_ids$, props = $labels|descriptions|aliases|claims$");
$data \leftarrow response$.json();
if id_only **then**
  $fetched_dict \leftarrow \emptyset$;
  foreach $id \in wikidata_ids$ **do**
    $label \leftarrow$ Extract English Label($data$["entities"][id]);
    $fetched_dict[id] \leftarrow label$;
  end
  return $fetched_dict$;
else
  $info_string \leftarrow$ "";
  foreach $id \in wikidata_ids$ **do**
    $entity \leftarrow data$["entities"][id];
    $label, desc, aliases \leftarrow$ Extract Basic Info($entity$);
    $info_string \leftarrow info_string + label, desc, aliases$;
    $properties \leftarrow$ get_entity_properties($id, entity$);
    foreach $prop \in properties$ **do**
      if *"entity"* $\in prop$ **then**
        $info_string \leftarrow info_string + prop$["entity"];
      else
        $info_string \leftarrow info_string + prop$["value"];
      end
    end
  end
  return $info_string$;
end

Algorithm 2: Get Entity Properties (`get_entity_properties`)

Input: $wikidata_id$, $entity$
Output: $selected_properties$
$claims \leftarrow entity$.get("claims");
$selected_props, ids_to_fetch \leftarrow \emptyset, \emptyset$;
$selected_props, ids_to_fetch \leftarrow$
fetch_property($claims, selected_props, ids_to_fetch$, needs_gold=True);
if *length*$(selected_props) < 10$ **then**
  $selected_props, ids_to_fetch \leftarrow$
  fetch_property($claims, selected_props, ids_to_fetch$, needs_gold=False);
end
if $ids_to_fetch \neq \emptyset$ **then**
  $fetched_ids \leftarrow$ get_wikidata_entity(list(ids_to_fetch), $id_only =$ True);
  foreach $prop \in selected_props$ **do**
    if $prop$.*get*(*"val_id"*) ***in*** ids_to_fetch **then**
      $prop$["entity"] $\leftarrow fetched_ids$.get($prop$["val_id"]);
    end
    if $prop$.*get*(*"property"*) ***in*** $fetched_ids$ **then**
      $prop$["property"] $\leftarrow fetched_ids[prop$["property"]];
    end
  end
end
return $selected_props$;

ID	Property	ID	Property
P31	instance of	P734	family name
P248	stated in	P361	part of
P17	country	P279	subclass of
P585	point in time	P276	location
P106	occupation	P1082	population
P21	sex or gender	P19	place of birth
P9259	epoch	P570	date of death
P569	date of birth	P69	educated at
P582	end time	P155	follows
P27	country of citizenship	P156	followed by

Table 2: Manually selected Wikidata Gold Properties utilized to prioritize high-value semantic relationships during knowledge retrieval.

System Prompt (LLM-as-a-Judge)

"You are an expert evaluator. Assess if the LLM Answer is correct based on the Short Answer (a Short Answer is a set of concise and correct answers).
You must respond only with a valid JSON object containing two keys: 'reasoning' (a brief explanation) and 'score' (integer 1 or 0):"
- reasoning: a brief explanation of why the LLM answer is correct or not based on the short answer.
- score: 1 if the LLM answer is correct, 0 otherwise.

Query: {query}
LLM Answer: {llm_answer}
Short Answer: {short_answer}*"*

As said in the Results section above, the LLM-as-a-Judge model's behaviour strictly penalizes valid answers, in the case in which they do not match the short answer word-for-word. This behaviour could be highly driven by the system prompt choice, which forces the model to keep in mind that the score heavily relies on whether the short answer is included or not: substantially, the model is largely conditioned by the given instructions.

A.4 Automatic Evaluation Metrics Results

Table 1 reports the comprehensive automated evaluation of all models across the three distinct pipelines (Baseline, RAG, and Oracle). To assess the impact of knowledge graph enrichment on the evaluation, each metric is split into two columns: *Standard* (computed against the original dataset's short answers) and *Wikidata* (computed against an expanded set of valid answers, retrieved via our custom Wikidata alias algorithm).

We know describe more in detail each result for both models. As said in the comparison paragraph, Baseline systems produce very low scores for each metric, while RAG and Oracle pipelines lead the models to a massive performance improvement. For instance, FLAN-T5's EM jumps from 0.0245 to 0.3475. The Oracle pipeline instead yields an

Algorithm 3: Fetch Selected Properties (`fetch_property`)

Input: $claims$, $selected_properties$, ids_to_fetch, $needs_gold$
Output: $selected_properties, ids_to_fetch$
foreach $prop_id, prop_value \in claims$ **do**
  if *length*$(selected_properties) \geq 10$ **then**
    break;
  end
  $is_gold \leftarrow prop_id \in$ GOLD_PROPERTIES;
  if $(needs_gold$ ***and*** $is_gold)$ ***or*** $($***not*** $needs_gold$ ***and not*** $is_gold)$ **then**
    $snak \leftarrow prop_value$[0]["mainsnak"];
    if $snak$*["snaktype"]* $==$ *"value"* **then**
      $datatype \leftarrow snak$["datatype"];
      if $datatype \in$ [*non-meaningful data types*] **then**
        continue;
      end
      $prop_name \leftarrow$ Get Gold Property Name$(prop_id)$;
      if ***not*** is_gold **then**
        Add $prop_id$ to ids_to_fetch;
      end
      $val_type \leftarrow snak$["datavalue"]["type"];
      $val_data \leftarrow snak$["datavalue"]["value"];
      if $val_type ==$ *"wikibase-entityid"* **then**
        $val_id \leftarrow val_data$["id"];
        Append {"property" : $prop_name$, "val_id" : val_id} to $selected_properties$;
        Add val_id to ids_to_fetch;
      else
        $formatted_data \leftarrow$ Extract amount, text, or time from val_data;
        Append {"property" : $prop_name$, "value" : $formatted_data$} to $selected_properties$;
      end
    end
  end
end
return $selected_properties, ids_to_fetch$;

Algorithm 4: Expand Ground Truth (`get_wikidata_ground_truth`)

Input: $wikidata_id$, $short_answer$
Output: $answers$ (Set of valid ground truth alternatives)
$answers \leftarrow \emptyset$;
Add $short_answer$.lower() to $answers$;
$response \leftarrow$ Call API(action = "wbgetentities", ids = $wikidata_id$, props = "claims");
$entity \leftarrow response$.json()["entities"][$wikidata_id$];
$properties \leftarrow$ get_entity_properties($wikidata_id, entity$);
$match \leftarrow$ Null;
foreach $prop \in properties$ **do**
    if *"entity"* $\in prop$ **then**
        $clean_answer \leftarrow short_answer$.strip().lower();
        $clean_prop \leftarrow prop$["entity"].strip().lower();
        if $clean_answer \subseteq clean_prop$ ***or*** $clean_prop \subseteq clean_answer$ **then**
            $match \leftarrow prop$["val_id"];
            break;
        end
    end
end
if $match \neq$ *Null* **then**
    $match_resp \leftarrow$ Call API(action = "wbgetentities", ids = $match$, props = "labels|aliases");
    $match_entity \leftarrow match_resp$.json()["entities"][$match$];
    $label \leftarrow$ Extract English Label($match_entity$);
    if $label \neq$ *Null and* $label \notin answers$ **then**
        Add $label$.lower() to $answers$;
    end
    $aliases \leftarrow$ Extract English Aliases($match_entity$);
    foreach $alias \in aliases$ **do**
        if $alias \notin answers$ **then**
            Add $alias$.lower() to $answers$;
        end
    end
end
return $answers$;

EM of 0.4250 for FLAN-T5. The gap of 0.0775 between RAG and Oracle demonstrates that our retriever is highly effective at fetching the relevant context.

A radically different behavior emerges between the two models, primarily driven by the verbosity bias. FLAN-T5 excels at outputting rigid, concise entities. Conversely, LLaMA is purely generative and outputs conversational, complete sentences. In the RAG setup, LLaMA reaches a subEM of 0.3525 against FLAN-T5's 0.4505, proving that LLaMA successfully extracts the correct information but embeds it in longer sequences.

Interestingly, METEOR scores reveal that, once context is provided, FLAN-T5 outclasses LLaMA (e.g., 0.5547 vs. 0.2804 on Oracle with Wikidata), showing that FLAN-T5 is natively calibrated for strict context exploitation.

Regarding the Wikidata improvement, the gain is particularly evident in the subEM metric. For the FLAN-T5 RAG pipeline, EM rises from 0.3475 to 0.3785, while subEM increases from 0.4505 to 0.5200. Similarly, LLaMA's subEM on RAG jumps from 0.3525 to 0.4410.

A.5 Manual Annotation Criteria

For the manual annotation phase, the two annotators decided on the following scoring criteria:

- **Correct Answer**: The answer is correct if:
  - The short (correct) answer is fully included in the model's answer;
  - The short (correct) answer is implicitly included in the model's answer;
  - The short (correct) answer is partially included in the model's answer, provided that it has been verified that it captures the full semantic domain of the short answer;
- **Incorrect Answer**: The answer is incorrect if:
  - The model's answer is not precise with respect to the short (correct) answer;
  - The model's answer does not include the short (correct) answer in any form;
  - The model's answer partially includes the short (correct) answer, provided that it has been verified that it does not capture the full semantic domain of the short answer;

To explain the partial correctness criteria, we give an explicit example: the query 'who used viruses to determine that DNA was the molecule of heredity' is associated to 'Alfred Hershey Martha Chase' as gold answer; if an evaluated model outputs 'Hershey and Chase', the prediction is scored as 'correct' as it successfully covers the full semantic domain of the gold answer.

A.6 Qualitative Analysis: Model Failures and Discoveries

Table 3 provides a detailed structural breakdown of specific queries extracted from the 250-sample qualitative review. This table illustrates the primary failure modes, the corrective impact of Wikidata, and the structural biases of the context navigation.

Table 3: Examples of generative model failures and behavioral breakthroughs, focusing on contextual confusion, semantic overlapping, order dependencies, and Wikidata retrieval dynamics.

Query ID	Query	Retrieved Context & Wikidata Information	Short Answer	Generated Answer
Category: Wikidata Response Rescue (Successful Fallback)				
PfDEgyXT7DwA	where is a sunday afternoon on the island of la grande jatte located	Document 1: The Island of la Grande Jatte is located at the very gates of Paris, lying in the Seine between Neuilly and Levallois-Perret... When Seurat began the painting in 1884, the island was a bucolic retreat far from the urban center. — Document 2: A Sunday Afternoon on the Island of La Grande Jatte... Seurat's composition includes a number of Parisians at a park on the banks of the River Seine. — Document 3: Seurat's painting was a mirror impression... For Parisians, Sunday was the day to escape the heat of the city... — Wikidata Information: A Sunday Afternoon on the Island of La Grande Jatte - painting by Georges Seurat, Art Institute of Chicago; instance of: painting; location: Art Institute of Chicago; collection: Art Institute of Chicago; inception: 1884-00-00.	Art Institute of Chicago	the Art Institute of Chicago
ks5AqfW3gIAN	unix command to copy a folder from one location to another	Document 1: Copy two files in the current directory into another directory: — Document 2: Copying a file to another file: — Document 3: Copying file(s) to a directory — Wikidata Information: cp - UNIX command for copying files and directories; instance of: standard UNIX utility or command; part of: GNU Core Utilities; has use: copying; cp; cp	cp	cp
JOkxTiFGfua7	when did the brown vs board of education end	Document 1: Brown v. Board of Education... Handed down on May 17, 1954, the Warren Court's unanimous decision stated... However, the decision's fourteen pages did not spell out any sort of method for ending racial segregation... — Document 2: The suit called for the school district to reverse its policy... — Document 3: The case was named after Oliver Brown as a legal strategy... — Wikidata Information: Brown v. Board of Education - 1954 U.S. Supreme Court case; point in time: 1954-05-17; court: Supreme Court of the United States.	1954-05-17	May 17, 1954
5yaLTAcNXmQG	where does the movie what if take place	Document 1: Wallace has an odd job and lives with his sister and nephew in Toronto, Ontario... — Document 2: Chantry is given a promotion to direct an animated project in Taiwan... learning that she has returned to Toronto... — Document 3: Over time the friendship grows and Wallace is invited to meet Ben... Ben later moves to Dublin, Ireland for six months... — Wikidata Information: The F Word - 2013 film by Michael Dowse; Aliases: What If; filming location: Toronto.	Toronto	Toronto, Ontario
Category: Contextual Order Bias and Document Navigation				

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Table 3 – continued from previous page

Query ID	Query	Retrieved Context & Wikidata Information	Short Answer	Generated Answer
krrAf0c84RJU	how much did germany have to pay in ww1	Document 1: During the period of reparations, Germany received between 27 and 38 billion marks in loans. By 1931, German foreign debt stood at 21.514 billion marks; ...With its defeat, Germany could not impose reparations and pay off her war debts now, which were now colossal. — Document 2: To pay towards this sum, Germany could pay in kind or in cash... German Government was to issue bonds at five per cent interest and set up a sinking fund of one per cent to support the payment of reparations. — Document 3: The Treaty of Versailles (signed in 1919) and the 1921 London Schedule of Payments required Germany to pay 132 billion gold marks (US$33 billion) in reparations to cover civilian damage caused during the war... — Wikidata Information: World War I reparations - war reparations; Aliases: None; instance of: war reparations; Commons category: German war reparations after World War I	132 billion gold marks	132 billion gold marks
AqPphDHmiar8	when did papua new guinea become an independent country	Document 1: ...administration began in 1906, although Papua remained under their control a British possession until the independence of Papua New Guinea in 1975. — Document 2: Papua New Guinea became self-governing on 1 December 1973 and achieved independence on 16 September 1975. — Document 3: Much of later economic policy, up to and beyond independence, proceeded. — Wikidata Information: history of Papua New Guinea - aspect of history; facet of: Papua New Guinea.	16 September 1975	16 September 1975
rt2JQvyAxBE9	who has home field advantage in the alds	Document 1: In Major League Baseball, the American League Division Series (ALDS) determines... — Document 2: Home field advantage goes to the team with the better regular season record... except for the wild card team, which never receives the home field advantage. — Document 3: Starting in 2007... MLB has decided to allow the team with the best record... — Wikidata Information: American League Division Series; sport: baseball.	Home field advantage goes to the team with the better regular season record	the team with the better regular season record
Category: Wikidata Failures (Low Context / Non-Direct Queries)				
Xh4ZSVbWmUXp	where is eden west resort from couples retreat	Document 1: ...they have found a couples therapy resort named Eden. — Document 2: The primary location for filming was at the St. Regis Bora Bora Resort, Bora Bora in French Polynesia. — Document 3: The retreat proves to be divided into Eden West and Eden East. West is for couples and uses the tagline "Stay Together". — Wikidata Information: Couples Retreat - 2009 film by Peter Billingsley; genre: romantic comedy; publication date: 2009-10-08.	Eden West is for couples and uses the tagline "Stay Together". East is for singles and uses the tagline "Come Together".	Bora Bora
bqvmCNmk3KfH	who was the oldest elected president of the united states of america	Document 1: The oldest living U.S. president is George H. W. Bush... — Document 2: The median age upon accession... the oldest was Donald Trump, who was 70 years, 220 days old at his inauguration... the oldest was Ronald Reagan, who was 73 years, 274 days old at the time of his election to a second term. — Document 3: On seven occasions, the oldest living president has acquired this distinction... — Wikidata Information: list of presidents of the United States by age - Wikimedia list article; is a list of: human.	John F. Kennedy	Donald Trump
1MjM2C2WghRm	what are the in the night garden characters	Document 1: The Tombliboos – Unn, Ooo, and Eee... The Tombliboos, along with Iggle Piggle, Upsy Daisy and Makka Pakka were first featured... — Document 2: The programme features a large cast of colourful characters with unusual names who live in a magical forest... — Document 3: The Ball is a colourful living ball that plays with the Night Garden characters. — Wikidata Information: In the Night Garden - television series; creator: Andrew Davenport; instance of: television series.	The Tombliboos – Unn (Andy Wareham), Ooo (Holly Denoon/Isaac Blake), and Eee (Elisa Laghi)	Iggle Piggle-Makka PakkaUpsy Daisythe Tombliboosthe Pontipines / the Wottingersthe Haahoosthe Ninky Nonkthe Pinky Ponk

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Table 3 – continued from previous page

Query ID	Query	Retrieved Context & Wikidata Information	Short Answer	Generated Answer
Category: Factual and Contextual Hallucinations (Multiple Entities Overlap)				
T1Kd9BKTQ6kz	2002 US Open men's champion	[Context provides overlapping accounts of both the men's and women's finals, featuring Pete Sampras and Serena Williams concurrently, causing severe semantic extraction failure]	Pete Sampras	Serena Williams
MCVlbR1Jn7UC	who made heard it through the grapevine first	[Documents contain multiple records tracking chronological releases and production involvements across multiple musical groups]	Norman Whitfield and Barrett Strong	Gladys Knight & the Pips

A.7 LLM-as-a-Judge and Human Annotation Discrepancies

Table 4 provides a detailed comparison for all queries where the LLM Judge (J) and human annotators (H) provided conflicting scores. This table contrasts the *Query*, the *Short Answer* (ground truth), and the model's *Generated Answer*.

References

Victor Wang and 1 others. 2025. Improving llm-as-a-judge inference with the judgment distribution.

Query ID	Query	Short Answer	Generated Answer	J	H
T1Kd9BKTQ6kz	who won the 2002 us open men's tennis championship	Pete Sampras	Serena Williams	1	0
T7FtET56pQ3T	where did frodo have to take the ring	Mount Doom	Sammath Naur	1	0
ks5AqfW3gIAN	unix command to copy a folder from one location to another	cp	cp	0	1
nCIMDOShVnvZ	what sports did they do in ancient greece olympics	foot race	The ancient Olympic Games were originally a festival, or celebration of and for Zeus; later, events such as a footrace, a javelin contest, and wrestling matches were added.	1	0
s0qUkNuXoOaM	what is the meaning of divergent in the movie	equal attributes of multiple factions	Her results show the rare quality of equal attributes of multiple factions	0	1
vQt10wCxKfLU	where was kenny chesney anything but mine filmed	Malibu, California and Myrtle Beach, South Carolina	Myrtle Beach	1	0
nLwLQf7ZoFzI	who wrote the first testament of the bible	Israelites	the Israelites	0	1
5AvhOqHxEAUn	where are most peanuts grown in the us	the southeastern United States region which includes Alabama, Georgia, and Floridathe southwestern United States region which includes New Mexico, Oklahoma, and Texasthe general eastern United States which include Virginia, North Carolina, and South Carolina	Georgia, Oklahoma, and Texas.	1	0
xaCXQyHzG76v	when does the playclock start in the nfl	the end of the previous down	In the NFL, teams have 40 seconds timed from the end of the previous down	1	0
mg4MlVVGNPDB	where was 20 million miles to earth filmed	Rome, Italythe U.S.	Rome	1	0
W8rl2NRPvr4u	what is make it stop by rise against about	It Gets Better campaign	The song explicitly deals with the bullying and harassment LGBT youth face.	0	1
ZWrbw4NnMzB5	who tried to steal the declaration of independence	Benjamin Franklin Gates	Ben	1	0
ykoi99oigSYB	town meetings practiced in many small new england towns are examples of what	direct democratic rule	open town meeting	1	0
O6EvTDzT5l2l	what was the war guilt clause in the treaty of versailles	the opening article of the reparations section of the Treaty of Versailles, which ended the First World War between the German Empire and the Allied and Associated Powers	Article 231	1	0
i0MgE6bfkcvS	in january of 1973 a ceasefire in vietnam was signed by all of the following except	United States Senate	The United States Senate	0	1
OF8p3CN56DUC	in javanese gamelan music typical elaborating instruments are	percussive	metallophones, xylophones, flutes, gongs, voices, as well as bowed and plucked strings.	0	1
nKfCTvqsshlL	iron pillar near qutub minar built by whom	king Chandra	king Chandra	0	1
ex4dPz1UsM0m	who is the song she'll be coming round the mountain about	the Second Coming of Christ and subsequent Rapture	the Second Coming of Christ	0	1
Y0Dm5eqqhP6C	who wrote the song you're welcome from moana	Lin-Manuel MirandaMark ManciniaOpetaia Foa'i	Lin-Manuel Miranda	1	0
7PGKeSaTh9hs	when was the last time the patriots went to the superbowl	2016–17	Super Bowl XLIX	1	0
chMKgYTh15tD	how many died in the las vegas shooting	58	60	1	0
lpudNDkqkQfe	what did the u.s. policy called the truman doctrine promise to do	created to counter Soviet geopolitical expansion during the Cold War	contain communism in Europe and elsewhere	0	1
VfwrBxvp6oaJ	who wrote manfred mann's blinded by the light	Bruce Springsteen	Bruce Springsteen	0	1
CaKpgYHIiI39	what did cro-magnon man paint on cave walls	animals	Cave paintings	1	0
afYpWNkwN12J	which is the characteristic of a line width depth color length	widthlength	A line has a width, direction, and length.	0	1
RHF22eWncpLG	where are voltage gated sodium channels concentrated in myelinated axons	plasma membrane	nodes of ranvier	1	0
cSVZv8wAsXKU	the national science day is observed every year on 28 february in honour of	Sir Chandrashekhara Venkata Raman	the discovery of the Raman effect	0	1
aUpMnjsWR52J	when did the new england patriots change their logo	2013	1993	0	1
VLAoKOGQIB0q	what type of poem is love among the ruins	rhyming couplets in which long trochaic lines are paired with short lines of three syllables	poem	1	0
Y8k0aXcqyT7X	how many cubes are in a 5x5x5 cube	98	There are 98 pieces on the exterior of the cube: 8 corners, 36 edges, and 54 centers (48 movable, 6 fixed).	0	1
rdsi6CUqvyT1	where is the baseball all-star game being played	July 11, 2017	Marlins Park	0	1

Table 4: Discrepancy matrix between LLM Judge (J) and Human Annotators (H) with full Query IDs.